

- (c) GSLV-F05
- (d) GSLV-F08
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

GSAT-6A is a high power S-band communication satellite of ISRO, which was launched by GSLV-F08 on 29 March, 2018 from SDSC SHAR, Sriharikota. Its mission life is 10 years.

183. In the context of space technology, what is 'BHUVAN', which is recently in the news ?

- (a) The name of the space vehicle proposed to be sent to the Moon to probe possible human habitation there.
- (b) A satellite launched by ISRO for promoting distant education in India.
- (c) A space telescope developed by India.
- (d) A geoportal of ISRO with 3-D imaging capabilities.

U.P. Lower Sub. (Pre) 2008

Ans. (d)

Bhuvan is a software application developed by ISRO, which allows users to explore a 2D/3D representation of the surface of the Earth. Apart from visualization it also provides disaster support services. It is an Indian version of Google Earth.

Defence Technology

Notes

IGMDP :

- The Integrated Guided Missile Development Programme (IGMDP) was conceived by renowned scientist **Dr. APJ Abdul Kalam** (then director of Defence Research and Development Laboratory – DRDL) to enable India to attain self-sufficiency in the field of missile technology.
- The IGMDP finally got the approval from the Govt. of India on 26 July, 1983. This ambitious project brought together the country's scientific community, academic institutions, R & D laboratories and industries in giving shapes to the strategic, indigenous missile system.
- The missiles developed under this programme are :
 - a. Short-range surface to surface ballistic missile – **Prithvi**
 - b. Intermediate-range surface to surface ballistic missile – **Agni**
 - c. Short-range low-level surface to air missile – **Trishul**
 - d. Medium-range surface to air missile – **Akash**
 - e. Third Generation anti-tank missile – **Nag**

- Dr. Kalam played a major role in the development and operationisation of Agni and Prithvi Missiles.

Prithvi Missile :

- Prithvi is a tactical surface to surface short-range ballistic missile developed by DRDO under the IGMDP. It is deployed by India's Strategic Forces Command.
- Its first test was conducted on 27 February, 1988 from (SHAR)- Sriharikota High Altitude Range.
- Its variants are as follows –
 - (a) Prithvi - I (ss-150) – Army version : 150 km range with a warhead of 1000 kg.
 - (b) Prithvi - II (ss-250) – Air Force version : 250-350 km range with a warhead of 500 kg.
 - (c) Prithvi - III (ss-350)/Dhanush – Naval version : 350 km range with a warhead of 1000 kg.
- Prithvi missiles are operated by liquid propellants.

Agni Missile :

- Agni missile came into existence after Prithvi missile, whose three tests were conducted on 22 May, 1989, 29 May, 1992 and 19 February, 1994 respectively from ITR Chandipur.
- It was double stage missile of which first stage was solid and second stage was liquid. So it can not be deployed in Military. It was only the technology demonstrator of Agni.
- Its advanced version – '**Agni-II**' was successfully test-fired on 11 April, 1999 from ITR Chandipur, which is capable to carry a warhead of 1000 kg and can strike target up to the distance of 2500 km.
- It is double stage missile which is 16 tonne heavy and 20 m long.
- For deployment in Military, solid propellant techniques are suitable because the solid propellants can kept filled before and can be fired.
- The missile can be fired just after getting any information.
- The height of **Agni-I** is 15 m and weight is 12 tonne.
- It can be fired up to a range of 700 km with a warhead of 1000 kg.
- It was firstly test-fired on 25 January, 2002 from ITR Chandipur.
- Agni - II and Agni - I have been deployed in Military.

Agni III :

- After maturation of Agni - II Technology, Agni - III came into existence.

- Its first successful test-fire was conducted on 12 April, 2007.
- It is capable to carry a warhead of 1500 kg. and can be fired up to a distance of 3000 km.

Agni IV :

- It is fourth in the Agni series of missiles.
- Its operational range is 4000 km.
- Its weight is 17 tonnes and length 20 metre.
- It is two-stage solid propellant missile.
- It is capable to carry a warhead weighing 1000 kg.

Agni V :

- Its first test was conducted on 19 April, 2012.
- Its length is 17.50 m and weight 50 tonne.
- It is three-stage missile.
- It is solid propellant operated missile and capable to carry both conventional and nuclear warhead of 1.5 tonne.
- Although its operational range is 5000 km, it can be extended up to 8000 to 10,000 km. It is the intercontinental ballistic missile (ICBM).
- On 15 December, 2022, first night trial of Agni V was successfully carried out by the Strategic Forces Command (SFC).

Agni P :

- Defence Research and Development Organisation (DRDO) successfully flight tested a New Generation Nuclear Capable Ballistic Missile Agni P (Agni Prime) from Dr. A.P.J. Abdul Kalam island off the coast of Odisha, Balasore on 28 June, 2021. It carried two multiple independently targetable reentry vehicles (MIRV).
- Agni P is a new generation advanced variant of Agni class of missiles.
- It is a two-stage canisterised solid propellant ballistic missile with dual redundant navigation and guidance system.
- It has maneuverable reentry vehicle (MaRV) capability that can deliver the warheads into two separate locations.
- It has range capability between 1,000 and 2,000 kms.
- On 18 December, 2021, the second flight test of Agni P was successfully conducted by the DRDO.
- On 21 October, 2022, Agni P was successfully test fired for the third time to achieve maximum range.

RudraM-I :

- The RudraM-I, previously referred as DRDO's anti-radiation missile (ARM) or new generation anti-radiation missile (NGARM) is an air-to-surface anti-radiation missile developed by the Defence Research and Development Organisation (DRDO).
- It is primarily meant for suppression of enemy air defenses (SEAD), that can be launched from a range of altitudes with large standoff distance for destroying enemy surveillance radars, tracking and communication systems.
- It is the first indigenous anti-radiation missile developed by India.
- It has a range of up to 200 km and it can be launched from altitudes of 500 m to 15 km.
- It was successfully flight tested on 9 October, 2020 by DRDO. It was launched from SU-30 MKI fighter aircraft on a radiation target located on Wheeler Island off the coast of Odisha.

RudraM-II :

- Defence Research and Development Organisation (DRDO) successfully flight-tested the RudraM-II air-to-surface missile from Su-30 MKI platform of the Indian Air Force (IAF) off the coast of Odisha on May 29, 2024.
- RudraM-II is an indigenously-developed solid-propelled air-launched missile system meant for Air-to-Surface role to neutralise many types of enemy assets.
- A number of state-of-the-art indigenous technologies developed by various DRDO laboratories have been incorporated in the missile system.
- It has a range of 300 km and has an additional Imaging Infrared (IIR) seeker.
- The missile can be launched from a variety of altitudes and has the capability to detect enemy radio frequencies and signals from radars at ranges exceeding 100 km.
- It is equipped with Lock-On-Before/After-Launch systems, providing flexibility in targeting.
- Additionally, the missile features an internal guidance system that enables it to autonomously navigate towards the target after launch.

Prahar Missile :

- Its weight is 1.2 Tonne with a length of 7.3 metre.
- It is single-stage solid propellant missile.
- It can be launched within 2-3 minutes without any preparation, providing significantly better reaction time

than liquid-fuelled Prithvi Ballistic Missile and act as a gap filler in the range between the Pinaca Multi-barrel Rocket Launcher and Smerch MBRL in due end and the Prithvi ballistic missile on the other.

- Rapid and the sharp reaction is the main peculiar feature of Prahar missile.
- It is capable to carry the warhead of 200 kg.
- It can strike target up to a distance of 150 km within 250 seconds.
- Prahar was first test-fired successfully on 21 July, 2011 from the Integrated Test Range (ITR) at Chandipur.

Pragati Missile :

- India's DRDO is promoting an indigenously developed short-range ballistic missile called Pragati, designed to strike targets up to ranges of 60-170 km.
- It is based on the Prahar missile.
- It is 7.4 m in length with 0.42 m diameter.
- It is capable to carry a conventional warhead of 200 kg.

Nirbhaya Cruise Missile :

- It is a long-range, all-weather, subsonic cruise missile designed and developed by DRDO.
- It can be launched from multiple platforms and is capable of carrying a conventional and nuclear warhead.
- It is a surface to surface missile.
- It is two-stage missile of long-range.
- It is six meter long, 0.52 m wide and with a wingspan of 2.7 m, it can carry the designated warhead at speed of 0.6-0.7 Mach.
- Its launch weight is about 1500 kg.
- It is remarkable that Nirbhaya missile is launched just like a rocket. Upon reaching the required velocity and height this turbofan engine missile takes over the further propulsion.
- Defence Experts considers Nirbhaya Missile equivalent to Tomahawk of U.S.A.

Amogh I Missile :

- Amogh I is a second generation, anti-tank, guided missile which has pinpoint accuracy for a range upto 2.8 km.
- It has been designed and developed by Bharat Dynamics Ltd. at Hyderabad.
- Its operational range is 2.8 km.
- The test firing was carried out on September 10, 2015 at Babina Army Range, Madhya Pradesh.

- In the test two missile were fired and both hit targets placed at 2.8 km as well as 2.6 km respectively.

K- Missile Family :

- The K-missile family named after Indian Scientist and former President A.P.J. Abdul Kalam, is a series of submarine-launched ballistic missile developed by India to boost its second-strike capabilities and thus augments its nuclear deterrence.
- First missile of this series is **Sagarika (K-15)** which has been tested successfully many times.
- Its launch platform is - Arihant Class Submarine.
- Its maximum firing range is 500 km.
- Its operational range is 3500 km.
- Its speed is 7.5 Mach.
- **Shaurya** is a land-based version of Sagarika K-15 Missile.

K-4 :

- It is a nuclear-capable intermediate-range submarine-launched ballistic missile under development by DRDO to arm the Arihant class submarines.
- This missile has a maximum range of 3500 km.
- It has been developed by DRDO under Black Project (A black project is a term used for a highly classified military project publically unacknowledged by the government, military personnel and contractors).

BrahMos Missile :

- The **BrahMos** is a medium-range ramjet supersonic cruise missile that can be launched from submarine, ships, aircraft, or land.
- It is a joint venture between the Russian Federation's NPO Mashinostroyeniya and India's Defence Research and Development Organisation (DRDO), who together have formed BrahMos Aerospace. The name BrahMos is a portmanteau formed from the names of two rivers, the Brahmaputra of India and the Moskva of Russia.
- BrahMos missile's original range was 290 km with speed of 2.8 to 3.0 Mach. In 2019, India upgraded the missile with a new range of 450/500 km. The 450/500 km variant of the BrahMos is referred to as BrahMos ER (Extended Range).
- A hypersonic version of the missile BrahMos-II, is also presently under development with a speed of Mach 7-8 to boost aerial fast strike capability.

- In February, 2024, the Cabinet Committee on Security has cleared the Indian Navy's proposals to buy 220 BrahMos ER supersonic cruise missiles for its warships.
- In 2016, as India became a member of the Missile Technology Control Regime (MTCR), India and Russia are now planning and jointly developing a new generation of BrahMos missiles with higher ranges (600 km, 800 km and 1500 km) and an ability to hit protected targets with pinpoint accuracy.
- In April, 2025, India has sent the second batch of BrahMos supersonic cruise missiles to Philippines. Earlier, in April, 2024, India delivered the first batch of BrahMos supersonic cruise missiles to Philippines. In January, 2022, Philippines concluded a \$375-million deal with India for three batteries of shore-based, anti-ship variant of the BrahMos becoming the first export customer for the joint venture missile between India and Russia.

Pralay :

- Defence Research and Development Organisation (DRDO) successfully conducted maiden flight test of indigenously developed surface-to-surface missile 'Pralay', from Dr. A.P.J. Abdul Kalam Island off the coast of Odisha on 22 December, 2021.
- **Pralay** is a canisterised surface-to-surface tactical short-range ballistic missile (SRBM) for battlefield use developed by the Defence Research and Development Organisation (DRDO).
- The missile is the amalgamation of technologies developed for exoatmospheric interceptor missile Prithvi Defence Vehicle (PDV) from Indian Ballistic Missile Defence Programme and Prahaar tactical missile.
- The Missile is powered with solid propellant rocket motor and many new technologies. The missile has a range of 150-500 kilometre and can be launched from a mobile launcher. The missile guidance system includes state-of-the-art navigation system and integrated avionics.
- On 23 December, 2021, DRDO conducted second test of Pralay from the Abdul Kalam Island. For the first time, two consecutive flight tests of a ballistic missile have been conducted successfully on two consecutive days. The second test concluded with a heavier payload and different range to check the precision and lethality of the missile.

Anti Ballistic Missile System :

- Indian Ballistic Missile Defence Programme is an initiative to develop and deploy a multi-layered ballistic missile defence system to protect from ballistic missile attacks.
- It is a double-tiered system consisting of two land and sea-based interceptor missiles namely the **Prithvi Air Defense (PAD)** missile for high altitude interception and the **Advanced Air Defense (AAD)** missile for lower altitude interception.
- The two-tiered shield could be able to intercept any incoming missile launched from 5000 km away.
- The system also includes an overlapping network of early warning and tracking RADAR, as well as command and control system.
- The PAD was first tested in Nov. 2006 followed by AAD in December, 2007.
- With the test of PAD, India became the 4th country to have successfully developed an anti-ballistic missile system after U.S.A., Russia and Israel.
- This anti-ballistic missile defence system has undergone several tests but it is yet to be officially commissioned.

Missile Technology Control Regime (MTCR) :

- The Missile Technology Control Regime (MTCR) is an informal political understanding among states that seek to limit the proliferation of missiles and missile technology.
- MTCR was formed in 1987 by the G-7 industrialized countries (Canada, France, Germany, Italy, Japan, the UK, and the United States).
- There are currently 35 countries that are members (Partners) of the MTCR : Argentina (1993); Australia (1990); Austria (1991); Belgium (1990); Brazil (1995); Bulgaria (2004); Canada (1987); Czech Republic (1998); Denmark (1990); Finland (1991); France (1987); Germany (1987); Greece (1992); Hungary (1993); Iceland (1993); India (2016); Ireland (1992); Italy (1987); Japan (1987); Luxemburg (1990); Netherlands (1990); New Zealand (1991); Norway (1990); Poland (1998); Portugal (1992); South Korea (2001); Russian Federation (1995); South Africa (1995); Spain (1990); Sweden (1991); Switzerland (1992); Türkiye (1997); Ukraine (1998); United Kingdom (1987); United States of America (1987). The date in brackets represents the initial year of membership.

- The MTCR seeks to limit the risks of proliferation of weapons of mass destruction (WMD) by controlling exports of goods and technologies that could make a contribution to delivery systems (other than manned aircraft) for such weapons. In this context, the Regime places particular focus on rockets and unmanned aerial vehicles capable of delivering a payload of at least 500 kg to a range of at least 300 km and on equipment, software, and technology for such systems.
- The MTCR is not a treaty and does not impose any legally binding obligations on Partners (members). Rather, it is an informal political understanding among states that seek to limit the proliferation of missiles and missile technology.
- The Australia Group is an informal forum of countries which, through the harmonisation of export controls, seeks to ensure that exports do not contribute to the development of chemical or biological weapons.
- The group, initially consisting of 15 members, held its first meeting in Brussels, Belgium, in June, 1985.
- With the incorporation of India on 19 January, 2018, it now has 43 members, including Australia, New Zealand, the European Union (represented by the European Commission), all 27 member States of the European Union, United Kingdom, the United States, Canada, India, Ukraine, and Argentina.
- The name comes from Australia's initiative to create the group. Australia manages the secretariat.
- All States participating in the Australia Group are parties to the Chemical Weapon Convention (CWC) and the Biological Weapon Convention (BWC).

The Hague Code of Conduct against Ballistic Missile Proliferation (HCoC) :

- The Hague Code of Conduct against Ballistic Missile Proliferation (HCoC) is the result of efforts of the international community to internationally regulate the area of ballistic missiles capable of carrying weapons of mass destruction.
- The Hague Code of Conduct (HCoC), was established on 25 November, 2002 as an arrangement to prevent the proliferation of ballistic missiles.
- The HCoC is the only multilateral code in the area of disarmament which has been adopted over the last years. The HCoC does not ban ballistic missiles, but it does call for restraint in their production, testing, and export.
- Along with the MTCR, the HCoC is the only multilateral transparency and confidence building instrument concerning the spread of ballistic missiles.
- Since the signing and entering into force of the HCoC Code in November, 2002 in The Hague, (Netherlands) the number of signatories has increased from 93 to 145 (143 UN members, the Cook Islands and the Holy See). In 2024, Qatar became the 145th subscribing State of the HCoC.
- India joined the HCoC on 1 June, 2016.

Australia Group :

- The **Australia Group** (AG) is a multilateral export control regime (MECR) and an informal group of countries (now joined by the European Commission) established in 1985 (after the use of chemical weapons by Iraq in 1984) to help member countries to identify those exports which need to be controlled so as not to contribute to the spread of chemical and biological weapons.
- **The Wassenaar Arrangement :**
- The Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies was formally established in 1996, replacing a previous export control mechanism (The Coordinating Committee on Multilateral Strategic Export Controls).
- **The Wassenaar Arrangement** is a multilateral export control regime (MECR) with 42 participating States.
- The 42 Participating States in the Wassenaar Arrangement are Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, the Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, India, Ireland, Italy, Japan, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Russia, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Türkiye, Ukraine, the United Kingdom, and the United States.
- The Wassenaar Arrangement promotes transparency, exchanges of views and information, and greater responsibility in transfers of conventional arms and dual-use goods and technologies with military applications.
- Participating States undertake to ensure that exports of these items do not contribute to the development or enhancement of military capabilities which undermine international and regional security and stability.
- Representatives of Participating States meet regularly in Vienna where the Wassenaar Arrangement's Secretariat is located.

New START (Strategic Arms Reduction Treaty) :

- New START (Strategic Arms Reduction Treaty) is a nuclear arms reduction treaty between the United States and the Russian Federation with the formal name of Measures for the Further Reduction and Limitation of Strategic Offensive Arms.
- It was signed on 8 April, 2010 in Prague, and, after ratification, entered into force on 5 February, 2011. It will now expire in February, 2026, having been extended in 2021.
- New START replaced the Treaty of Moscow (SORT), which was to expire in December, 2012.
- It follows the START I treaty, which expired in December 2009; the proposed START II treaty, which never entered into force; and the START III treaty, for which negotiations were never concluded.
- The treaty calls for halving the number of strategic nuclear missile launchers and establishing a new inspection and verification regime, replacing the SORT mechanism. It does not limit the number of operationally inactive nuclear warheads that can be stockpiled, a number in the high thousands.
- On 21 February, 2023, Russia suspended its participation in New START. However, it did not withdraw from the treaty, and clarified that it would continue to abide by the numerical limits in the treaty.

The North Atlantic Treaty Organization (NATO) :

- The North Atlantic Treaty Organization (NATO), also called the North Atlantic Alliance, is an intergovernmental military alliance of 32 member States – 30 European and 2 North American.
- Established in the aftermath of World War II, the organization implements the North Atlantic Treaty, signed in Washington, D.C., on 4 April, 1949.
- NATO is a collective security system : its independent member states agree to defend each other against attacks by third parties.
- NATO formed with twelve founding members and has added new members ten times, most recently when Sweden joined the alliance on 7 March, 2024. In addition, NATO currently recognizes Bosnia and Herzegovina, Georgia, and Ukraine as aspiring members.
- NATO's main headquarters are located in Brussels, Belgium, while NATO's military headquarters are near Mons, Belgium.

Question Bank

1. The motto of Defence Research and Development Organisation is :

- (a) Jagratam Aharnisham
- (b) Tejasvinavdhitamastu
- (c) Balasya Mulam Vigyanam
- (d) Vayam Rakshamah

R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

The Defence Research and Development Organisation (DRDO) is the premier agency under the Department of Defence Research and Development in Ministry of Defence of the Government of India, charged with the military's research and development, headquartered in Delhi, India. DRDO was formed in 1958 from the amalgamation of the then already functioning Technical Development Establishment (TDEs) of the Indian Army and the Directorate of Technical Development and Production (DTDP) with the Defence Science Organization (DSO). The motto of DRDO is : "Balasya Mulam Vigyanam" – the source of strength is science – which drives the nation in peace and war.

2. Where did the Indian Army establish a Quantum Laboratory in Madhya Pradesh in 2021?

- (a) Mhow
- (b) Jabalpur
- (c) Bhopal
- (d) Chhindwara

M.P. P.C.S. (Pre) 2022

Ans. (a)

Indian armed forces are doing dedicated investment in quantum technologies for some time now. During 2021, Indian Army set up a Quantum Computing Laboratory at Military College of Telecommunication Engineering, Mhow, Madhya Pradesh. This laboratory has been established with the help of the National Security Council Secretariat (NSCS). Research undertaken by Indian Army in the field of Quantum Technology will help leapfrog into next generation communication and transform the current system of cryptography in the Indian Armed Forces to Post Quantum Cryptography (PQC).

3. Mission 'Divyastra' is related to which of the following?

- (a) MIRV Technology
- (b) Prithvi Missile System
- (c) Mounted Gun System
- (d) Reconnaissance Vehicle (wheeled)

Uttarakhand P.C.S. (Pre) 2024

Ans. (a)

Mission Divyastra, the first flight test of indigenously developed Agni-V missile with Multiple Independently Targetable Re-entry Vehicle (MIRV) technology was successfully conducted by the Defence Research and Development Organisation (DRDO) from Dr. APJ Abdul Kalam Island in Odisha on 11 March, 2024. Various Telemetry and radar stations tracked and monitored multiple re-entry vehicles. The Mission accomplished the designed parameters. MIRV is an exoatmospheric ballistic missile payload containing several warheads, each capable of being aimed to hit a different target.

4. What is the name of the manportable air defence system developed by 'DRDO' to neutralize aerial threats?

(a) Akash-NG (b) QRSAM
(c) VL-SRSAM (d) VSHORADS

Rajasthan P.C.S. (Pre) 2024

Ans. (d)

The Very Short-Range Air Defence System, or VSHORADS, is a 4th Generation, technically-advanced miniaturized, man-portable air-defence system (MANPADS) developed by Research Centre Imarat (RCI), a research lab under Defence Research and Development Organisation (DRDO), located in Hyderabad. Multiple DRDO laboratories along with Indian industry partners are participating in the project. It is designed for anti-aircraft warfare and neutralising low altitude aerial threats at short ranges. Recently, on February 1, 2025, DRDO has successfully conducted three successive flight-trials of VSHORADS from Chandipur off the coast of Odisha. Earlier, DRDO successfully conducted three flight-tests of VSHORADS at the Pokhran Field Firing Ranges in Rajasthan on October 3 and 4, 2024.

5. The name of India's first indigenous anti-radiation missile is :

(a) Tandav (b) Trinetra
(c) Saksham (d) Rudram
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

A New Generation Anti-Radiation Missile (NGARM), RudraM-I, was successfully flight-tested on 9th October, 2020 by Defence Research and Development Organization (DRDO). It was launched from SU-30 MKI fighter aircraft onto a radiation target located on Wheeler Island off the coast of Odisha. This is the first indigenous anti-radiation air-to-surface missile of India. It has a range of up to 200 km and it can be launched from altitudes of 500 m to 15 km. An anti-

radiation missile can locate and target any radiation - emitting source like enemy radars, communication sites and other Radio Frequency (RF) emitting targets.

6. With reference to RudraM-II Missile, which of the following statements is/are correct?

1. It is an air-to-air missile.
2. It was tested in Rajasthan in May, 2024.

Select the correct answer using the codes given below:

(a) Only 2 (b) Neither 1 nor 2
(c) Only 1 (d) Both 1 and 2

U.P.P.C.S. (Pre) 2024

Ans. (b)

The RudraM-II is an indigenously developed, solid-propelled, air-to-surface anti-radiation missile system developed by DRDO, successfully flight-tested from a Su-30 MKI fighter jet off the coast of Odisha on May 29, 2024. The RudraM-II is designed for air-to-surface operations, specifically to neutralize various types of enemy assets, including ground radars and communication stations. A number of state-of-the-art indigenous technologies developed by various DRDO laboratories have been incorporated in the missile system. Hence, both statements are incorrect.

7. India's Anti tank missile is-

(a) Nag (b) Agni
(c) Vajra (d) Aakash

M.P. P.C.S. (Pre) 1991

M.P. P.C.S. (Pre) 1990

Ans. (a)

'Nag' is a "Fire and Forget" anti tank missile developed in India. Land based Nag Missile has a range of 500 m to 4 Km. It's first successful test was done on November 24, 1990. It uses nitramine-based smokeless extruded double base sustainer propellant.

8. Match List-I with List-II and select the correct answer using the codes given below.

List-I	List-II
A. Prithvi	1. Multi-barrel rocket launcher
B. Trishul	2. A remote operated battlefield inspection vehicle
C. Pinaka	3. Surface-to-surface missile
D. Nishant	4. Surface to Air Missile

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	1	2
(c)	3	4	2	1
(d)	2	3	4	1

U.P. Lower Sub. (Pre) 2002

Ans. (b)

Prithvi – Surface-to-surface missile
 Trishul – Surface-to-air missile
 Pinaka – Multi-barrel rocket launcher
 Nishant – A remote operated battlefield inspection vehicle

9. Which of the following is not a missile?

- (a) Agni (b) Akash
 (c) Pinaka (d) Nag

M.P.P.C.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

10. India's surface to air missile is –

- (a) Agni (b) Nag
 (c) Trishul (d) Prithvi

R.A.S./R.T.S.(Pre) 2013

Ans. (c)

Trishul is a short range surface to air missile developed by India as a part of the Integrated Guided Missile Development program. Prithvi and Agni are tactical surface to surface short range ballistic missiles, while Nag is a anti-tank missile.

11. Which one of the following is correctly matched?

- (a) Trishul – Surface to surface missile
 (b) Prithvi – Surface to air missile
 (c) NAG – Anti-tank missile
 (d) Pinaka – Light combat aircraft

U.P. U.D.A./L.D.A. (Pre) 2013

U.P.P.C.S. (Pre) 2012

Ans. (c)

See the explanation of above question.

12. Which of the following is India's surface to surface defence missile –

- (a) Nag (b) Trishul
 (c) Akash (d) Prithvi

U.P.P.C.S. (Pre) 1990

Ans. (d)

Prithvi is India's surface to surface defence missile while Akash is a medium-range mobile surface to air missile defence system.

13. Match List I with List II and select the correct answer using the code given below the lists :

- | List I | List II |
|------------|--|
| A. Trishul | 1. Anti tank missile |
| B. Prithvi | 2. Intermediate range ballistic system |
| C. Agni | 3. Short range surface to air missile |

D. NAG

4. Surface to surface missile

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 2 | 1 | 4 | 3 |

U.P.U.D.A./L.D.A. (Pre) 2007

U.P.P.C.S. (Pre) 2001

Ans. (c)

Trishul – Short range surface to air missile
 Prithvi – Surface to surface missile
 Agni – Intermediate range ballistic system
 NAG – Anti-tank missile

14. Match the following :

List - I

List - II

- | | |
|------------------------|---------------------------------|
| A. Akash | 1. Kanshiram |
| B. Bahujan Samaj Party | 2. A surface-to-air missile |
| C. Prithvi | 3. A surface-to-surface missile |
| D. Mirage 2000 | 4. Fighter plane |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 4 | 2 | 1 |
| (c) | 1 | 3 | 4 | 2 |
| (d) | 2 | 1 | 3 | 4 |

M.P.P.C.S. (Pre) 1994

Ans. (d)

Akash is a medium range surface-to-air missile. Kashiram was the founder of Bahujan Samaj Party. Prithvi is a surface to surface missile. Mirage 2000 is a French multirole single engine fourth-generation jet fighter.

15. Which of the following statements are true about 'Nirbhay' Missile, which was successfully flight-tested on 7th November, 2017?

- It can deliver nuclear warheads to a strike range of 1000 km.
- It is a land-attack cruise missile.
- It is an effective counter to Pakistan's 'Babur' missile.
- It is a supersonic missile.

Select the correct answer from the codes given below :

- (a) Only 1 and 2 (b) Only 1, 2 and 3
 (c) Only 2, 3 and 4 (d) All 1, 2, 3 and 4

U.P.P.S.C. (GIC) 2017

Ans. (b)

On 7th November, 2017 Defence Research and Development Organisation (DRDO) successfully test-fired its first indigenously designed and developed long-range subsonic cruise missile 'Nirbhay' from Integrated Test Range (ITR) at Chandipur (Odisha). It is a land-attack cruise missile which can deliver nuclear or conventional warheads of up to 300 kg to a strike range of 1000 km. Defence experts considers Nirbhay missile equivalent to Tomahawk of U.S.A. It is an effective counter to Pakistan's 'Babur' missile which is a medium-range (450-750 km) subsonic cruise missile.

16. Which one of the following statements is not correct about Astra missile ?

- (a) It is an air-to-air ballistic missile.
- (b) It has been indigenously built by The DRDO
- (c) It was successfully test-fired on 20 May, 2011 from Chandipur
- (d) It has the capacity to destroy the enemy plane up to a distance of 90 km.

U.P.P.C.S. (Mains) 2010

Ans. (d)

Astra is an active radar homing Beyond-Visual-Range Air-to-Air Missile (BVRAAM) developed by the DRDO. It has the capacity to destroy the enemy plane up to a distance of 80 km. Ballistic flight test of the missile was carried out from the integrated test range at Chandipur on 20 May, 2011.

17. India's first Beyond-Visual Range (BVR) air-to-air missile indigenously designed & developed by Defence Research & Development Organization (DRDO) is :

- (a) Prithvi
- (b) Agni
- (c) Astra
- (d) Akash

R.A.S./R.T.S. (Pre) 2018

Ans. (c)

Astra is an all-weather beyond visual range air-to-air missile (BVRAAM) developed by the Defence Research and Development Organisation. It is the first air-to-air missile developed by India. Astra is designed to be capable of engaging targets of varying range and altitude allowing for engagement of both short-range targets at a distance of 20 km and long-range targets up to a distance of 80 km.

18. Missile 'Astra' is a –

- (a) Land to land missile
- (b) Land to air missile
- (c) Air to air missile
- (d) Water to land missile

U.P.P.C.S (Pre) 2011

Ans. (c)

See the explanation of above question.

19. On 23rd August 2023, LCA (Light Combat Aircraft) Tejas successfully test fired ASTRA missile, this missile is :

- (a) Beyond Visual Range Surface to Air Missile
- (b) Beyond Visual Range Air to Air Missile
- (c) Within Visual Range Air to Air Missile
- (d) Within Visual Range Surface to Air Missile

Rajasthan P.C.S. (Pre) 2023

Ans. (b)

Light Combat Aircraft (LCA) Tejas, Limited Series Production-7 (LSP-7) successfully test fired the ASTRA indigenous beyond visual range air-to-air missile off the coast of Goa on 23 August, 2023. The missile release was successfully carried out from the aircraft at an altitude of about 20,000 feet and all the objectives of the test were achieved with a perfect launch.

20. The name of successfully tested air-to-air missile in Balasore, in May 2003 is :

- (a) Astra
- (b) Akash
- (c) Tejas
- (d) Vajra

U.P. Lower Sub. (Pre) 2004

Ans. (a)

Astra is an active radar homing beyond-visual-range air-to-air missile developed by DRDO. The missile was first successfully tested on 9 May, 2003.

21. Which missile programme has been shelved by the DRDO?

- (a) Nag
- (b) Trishul
- (c) Agni
- (d) None of these

Chhattisgarh P.C.S. (Pre) 2003

Ans. (b)

Trishul, a supersonic missile programme, has been shelved by DRDO in 2008 due to some technical flaws.

22. The short-range missile – TRISHUL, test fired from Chandipur is a :

- (a) Subsonic missile
- (b) Supersonic missile
- (c) Hypersonic missile
- (d) None of these

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

See the explanation of above question.

23. India's short-range missile is –

- (a) Agni
- (b) Arjun
- (c) Nag
- (d) Prithvi

U.P.P.C.S. (Pre) 1996

Ans. (c)

Nag missile has the lowest-range in the given options. It's land based version has a range of maximum 4 km. Prithvi-I missile has a minimum range of 150 km. Among the different versions of the Agni missiles, Agni-1 has a minimum range of 700 km. Arjun is the third generation main battle tank developed by the DRDO.

24. The name 'Agni' is related with –

- (a) A machine to fight with fire
- (b) A warfare tank
- (c) An aeroplane
- (d) The missile developed by India

42nd B.P.S.C. (Pre) 1997

Ans. (d)

In 1980's the Integrated Guided Missile Development Programme (IGMDP) was conceived by renowned scientist Dr. APJ Abdul Kalam and under this programme AGNI, an intermediate range surface-to-surface ballistic missile was developed.

25. Which one of the following statements is incorrect ?

- (a) INFANT-82 is Indian Naval Fast Attack Craft.
- (b) A virus cannot reduce environmental pollution.
- (c) Typha is a computer virus.
- (d) Brahmos is a supersonic cruise missile.

U.P.P.C.S. (Mains) 2004

Ans. (b)

INFANT-82 is Indian Naval's Fast Attack Craft which was inducted in Indian Navy on 29 October, 2003. It was built with the help of Israel. Some of the virus positively affect the environment and helps to reduce the pollution. Brahmos is a supersonic cruise missile. It is a joint venture between the Russian Federation and India. Typha is a computer virus. So, from the above statements option (b) is incorrect.

26. In the context of Indian defence which one of following statements is NOT correct ?

- (a) With the induction of Prithvi-II, the IAF is the only Air Force in the world with surface to surface ballistic missiles under its command.
- (b) Sukhoi-30 MKI jet fighters can launch Air-to-Air and Air-to-Surface precision missiles.
- (c) Trishul is a supersonic Surface to Air missile with a range of 30 Km.
- (d) The indigenously built INS Prabal can launch Surface to Surface missiles.

I.A.S. (Pre) 2003

Ans. (c)

Statement of option (c) is incorrect. Trishul was a short range surface-to-air missile developed by India as a part of IGMD programme. It had a range of about 9 km. In 2008, India had officially announced the shut down of this project. The rest of the statements are correct.

27. Consider the following statements :

- 1. In November, 2006, DRDO successfully conducted the interception test using a Prithvi-II missile.
- 2. Prithvi-II is a surface to surface missile and can be deployed to guard the metros against air attacks.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (a)

Prithvi-II is a single-stage liquid fuelled missile having a maximum warhead mounting capability of 500 kg with a range of 250 Km. In November, 2006 DRDO successfully conducted the interception test using a Prithvi-II missile. In this test an exo-atmospheric hypersonic interceptor missile was destroyed an 'enemy' Prithvi missile at an altitude of 50 km. In this test Prithvi-II was not for guard/protection but an attacking missile. The exo-atmospheric hypersonic interceptor missiles of Prithvi Air Defence (PAD) can be deployed to guard the metros against air attack.

28. Consider the following aircraft:

- 1. Rafael
- 2. MiG-29
- 3. Tejas MK-1

How many of the above are considered fifth generation fighter aircraft?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

I.A.S. (Pre) 2024

Ans. (d)

Fifth generation fighter aircrafts include major technologies developed during the first part of the 21st century. As of 2024, these are the most advanced fighters in operation. The characteristics of a fifth generation fighter aircraft are not universally agreed upon, and not every fifth generation type necessarily has them all; however, they typically include stealth, low-probability-of-intercept radar (LPIR), agile airframes with supercruise performance, advanced avionics features, and highly integrated computer systems capable of networking with other elements within the battlespace for situational awareness and C³ (command, control and communications) capabilities. Examples of combat-ready

fifth generation fighter aircrafts include the Lockheed Martin F-22 Raptor and Lockheed Martin F-35 Lightning II (USA), Chengdu J-20 (China), and Sukhoi Su-57(Russia). Rafael is referred to as an 'omnirole' 4.5th generation aircraft developed by France's Dassault Aviation. The MiG-29 is a 4th generation fighter jet developed by Russia (Former Soviet Union). The Tejas MK-1 is an indigenously designed and developed multirole Indian fighter jet, but it too falls under the 4.5th generation. Hence, option (d) is the correct answer.

29. C-130J Super Hercules is world's most advanced air lifter being used by Indian Air Force. These air lifters (Air-craft) are made by –

- (a) Israel (b) U.S.A.
(c) Russia (d) France

R.A.S/R.T.S. (Pre) 2013

Ans. (b)

C-130J Super Hercules is world's most advanced air lifter being used by Indian Air Force. IAF had signed a deal for about 1.1 billion Dollars with the US Military Systems giant Lockheed Martin on January, 2008. The Lockheed Martin Aeronautics is headquartered in Fort Worth, Texas (U.S.A.).

30. Which of the following aircraft is an air-to-air refueller of the Indian Air Force?

- (a) Ilyushin IL-76 (b) C-130 J Super Hercules
(c) C-17 Globemaster III (d) Ilyushin IL-78

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

The Ilyushin IL-78 is a Soviet four-engined aerial refuelling tanker. The Indian Air Force refers to the aircraft as 'MARS' (Mid-Air Refuelling System) and has raised a new unit (No. 78 sqn).

31. Which one of the following is not an aircraft ?

- (a) Jaguar (b) Dornier-228
(c) Sarath (BMP-II) (d) MIG-27M

U.P.P.C.S. (Mains) 2009

Ans. (c)

BMP-II is an amphibious infantry fighting vehicle introduced in the 1980s in the Soviet Union. The Indian version of BMP-II is known as Sarath.

32. Match List-I with List-II and select the correct answer using the code given below the list :

- | List-I | List-II |
|------------|------------------------------|
| A. Arihant | 1. American Cryogenic Rocket |
| B. Awacs | 2. Anti-tank missile |

C. Atlas Centaur 3. Israel developed Falcon for Air Force

D. Nag

4. Atomic submarine

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 3 | 2 | 1 |
| (b) | 4 | 3 | 1 | 2 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 4 | 2 | 1 |

U.P.P.C.S (Pre) 2010

Ans. (b)

INS Arihant is India's nuclear-powered submarine. It was built under the Advanced Technology Vessel (ATV) Project. Awacs (Airborne Warning and Control system) is an airborne radar picket system designed and developed by Israel Aerospace Industries. Atlas Centaur is an American cryogenic rocket. It was the first rocket which used the cryogenic engine. Nag is a third generation anti-tank missile developed in India.

33. Which among the following is a nuclear-powered submarine of Indian Navy?

- (a) INS Shalki (b) INS Sindhuvir
(c) INS Shishumar (d) INS Chakra

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

INS Chakra is an 8,140 ton Akula class, nuclear-powered submarine. It was commissioned on 4th April, 2012.

34. Which of the following is a submarine of Indian Navy?

- (a) INS Viraat (b) INS Sindhurakshak
(c) INS Rajali (d) INS Vikrant

M.P. P.C.S. (Pre) 2016

Ans. (b)

INS Sindhurakshak was a Russian-made diesel-electric submarine of the Indian Navy. The submarine suffered a minor fire incident in 2010 and a major one on 14 August, 2013, which resulted in its sinking at Mumbai's naval dockyard.

35. India's first armed nuclear submarine is

- (a) I.N.S. Tihayu (b) I.N.S. Virat
(c) I.N.S. Arihant (d) I.N.S. Astradharini

Jharkhand P.C.S. (Pre) 2016

Ans. (c)

Admiral Sunil Lamba formally inducted India's first indigenously built nuclear submarine INS Arihant in Indian Navy in August, 2016. It is operated by a 83 MW pressurized light nuclear reactor. It weighs 6000 tons. India is the 6th country in the world to achieve nuclear trilateral capacity for atomic warfare after land and sky.

36. The effective range of Agni-I missile test-fired in July 2004, is –

- (a) Above 700 km but less than 2000 km
- (b) 300 km
- (c) 400 km
- (d) 2500 km

U.P.P.C.S. (Mains) 2003

Ans. (a)

Agni-I is a surface to the surface short-range ballistic missile. It has an effective range of 700-1250 km.

37. An indigenously manufactured missile with a range of 700 km was successfully tested by India in March, 2010. How is it known ?

- (a) Prithvi - I
- (b) Prithvi - II
- (c) Agni - I
- (d) Agni - II

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c)

A nuclear capable surface-to-surface Agni-I ballistic missile with a range of 700 km was successfully tested from the Wheeler's Island off the Odisha Coast on 28 March, 2010. Its first trial in the night was successfully completed on 11 April, 2014.

38. What is the range of 'Dhanush' missile tested successfully by India in March, 2010 ?

- (a) 250 km
- (b) 300 km
- (c) 350 km
- (d) 400 km

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c)

Dhanush missile is the naval version of Prithvi missile. It has a range of 350 Km.

39. The ICBM developed by India, with a strike range of more than 2000 km, is given the name :

- (a) Prithvi
- (b) Trishul
- (c) Akash
- (d) Agni-II

U.P.P.C.S. (Mains) 2011

U.P.P.C.S. (Mains) 2006

Ans. (d)

India has developed a series of AGNI missiles under the Integrated Guided Missile Development Programme (IGMDP). AGNI-I is a short-range ballistic missile with an effective range of 700-1250 km, while AGNI-II has an effective range of 2000-2500 km.

40. Which one of the following is named 'Hansa-2'?

- (a) A Satellite
- (b) A Trainer Aircraft

- (c) A Submarine
- (d) Surface to Surface Missile

U.P.P.C.S. (Pre) 1996

Ans. (b)

Hansa is a general aviation airplane for flight training, designed by the National Aerospace Laboratories (NAL) of India and built by Taneja Aerospace and Aviation Limited (TAAL). Some different versions of Hansa Aircraft are Hansa-2, Hansa-2RE, Hansa-3, Hansa-S and Hansa-UAV.

41. Which one of the following pair is not correctly matched?

- (a) Arjun : Indigenously produced Main Battle Tank (MBT)
- (b) Phalcon : Cruise missile supplied by Russia to India
- (c) Saras : Indigenously developed civilian passenger Aircraft
- (d) Operation Seabird : New Indian naval base at Karwar

I.A.S. (Pre) 2005

Ans. (b)

The Phalcon is an Israel manufactured Airborne Early Warning Command and Control (AEWC&C) system. It provides real time surveillance of a few hundred kilometres of territory and also command and control. It can pick up a low-flying aircraft, a missile and provide an advance warning after correlation. India bought this system from Israel. Other pairs are correctly matched.

42. From which country of the world will Indian obtain 'Phalcon Radar' system ?

- (a) United States of America
- (b) Israel
- (c) France
- (d) Russia

U.P.P.C.S. (Mains) 2004

Ans. (b)

See the explanation of above question.

43. From which one of the following did India buy the Barak anti-missile defence systems?

- (a) Israel
- (b) France
- (c) Russia
- (d) USA

I.A.S. (Pre) 2008

Ans. (a)

India bought the Barak anti-missile defence system from Israel. The Barak missiles were firstly deployed on India's aircraft carrier INS Virat.

44. Barak-8 Missile (next generation) has been developed by India in collaboration with which of the following countries?

- (a) Russia
- (b) U.S.A.

(c) Israel

(d) France

U.P. P.C.S. (Pre) 2016

Ans. (c)

Barak-8 (the Hebrew word for Lightning) is an Indian-Israeli surface-to-air missile (SAM), designed to defend against any type of airborne threat including aircraft, helicopters, anti ship missiles, and UAVs as well as cruise missiles and combat Jets.

45. With the co-operation of which one of the following countries, a pilotless helicopter is being developed for Indian Navy ?

(a) China

(b) France

(c) Israel

(d) U.S.A.

U.P.P.C.S. (Mains) 2008

Ans. (c)

The unmanned helicopter is jointly developed by Hindustan Aeronautics and Israel Aircraft Industries.

46. Which French company supplied Rafale fighter jets to India?

(a) LH Aviation

(b) Dassault Aviation

(c) Isoire Aviation

(d) Humbert Aviation

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

French company Dassault Aviation supplied Rafale fighter jets to India. They delivered 36 'Rafale' fighter jets to the Indian Air Force (IAF). IAF received the delivery of the 36th Rafale in December, 2022, completing the deal signed in 2016.

47. Which of the following is a modern tank?

(a) Bhim

(b) Akash

(c) Arjun

(d) Prithvi

I.A.S. (Pre) 1996

Ans. (c)

Arjun is an extremely subversive modern main battle tank developed by DRDO with the help of Bharat Heavy Electricals Ltd., Bharat Earth Movers Limited and Heavy Vehicles Factory (Tamil Nadu).

48. The first indigenously assembled T-90 S battle tank has been named :

(a) Arjun

(b) Bhima

(c) Bhishma

(d) Tarang

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

The first indigenously assembled T-90S battle tank has been named 'Bhishma'.

49. Consider the following statements in context to Agni (II) missile which was launched in space on January 25, 2002.

1. It is a surface-to-surface missile with a range of 700 km.

2. It was launched at 8:45 am in space from Wheeler's Island, Orissa.

3. It is a two-stage missile whose first phase has a liquid propellant rocket motor and in second, has a solid propellant rocket motor.

(a) Only 1

(b) 1 and 2

(c) 2 and 3

(d) 1 and 3

U.P. Lower Sub. (Spl.) (Pre) 2002

Ans. (*)

Agni-II is a medium range ballistic missile with a range of 2000-2500 km. These missiles use solid propellant in its two stages. Nuclear capable AGNI-1 missile was tested by DRDO at the Interim Test range in Chandipur on 25 January, 2002 at 8:50 am not AGNI-II. AGNI-I is a solid-fuel first stage missile with a range of 700-1250 Km.

50. Agni-II missile was launched from :

(a) Sriharikota

(b) Chandipur

(c) Thumba

(d) Pokharan

U.P.P.C.S. (Mains) 2004

Ans. (b)

Agni-II, a surface-to-surface medium-range ballistic missile was test fired from the Wheeler's Island near the Interim test range at Chandipur, Odisha on 17 January, 2001. It has a range of 2000-2500 km. India on 16 November, 2019 conducted successfully the first night trial of Agni-II.

51. The approx. range of Agni-II missile is –

(a) 500 km

(b) 2000 km.

(c) 3500 km

(d) 5000 km.

I.A.S. (Pre) 2001

Ans. (b)

See the explanation of above question.

52. With reference to Agni-IV Missile, which of the following statements is/are correct?

1. It is a surface-to-surface missile.

2. It is fuelled by liquid propellant only.

3. It can deliver one-tonne nuclear warheads about 7500 km away.

Select the correct answer using the code given below.

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2014

Ans. (a)

Agni-IV is the fourth in the Agni series of missiles which was earlier known as Agni-II Prime. It is to be noted that Agni missiles are long range, nuclear weapon capable surface to surface ballistic missiles. Agni-IV has two stages of solid propulsion and a payload with re-entry heat shield. Agni-IV can take a conventional or nuclear warhead of 1 tonne to about 4000 km range nuclear capable ballistic missile. So, only statement 1 is correct.

53. Which one of the following facts about Agni-5 missile is not correct ?

- (a) It can be fired from any place.
(b) It has a firing range of up to 5000 km.
(c) It cannot be captured by Radar of enemies.
(d) It is not exclusively indigenous.

U.P.P.C.S. (Pre) 2015

Ans. (d)

The Agni-V missile, having a range of 5,000 km is a fully indigenous missile. Therefore option (d) is incorrect. Agni-V is about 17 metres long, 2 metres wide and has a launch weight of around 50 tons. The missile can carry a nuclear warhead of more than one ton.

54. 'INDRA' is –

- (a) Missile (b) Unmanned aircraft
(c) Radar (d) Tank

U.P.Lower Sub. (Pre) 2004

U.P.Lower Sub. (Pre) 2003

Ans. (c)

INDRA is a series of 2D radars meant for Army and Air Force. This was the first high power radar developed by DRDO. This system is also used by Sri Lanka.

55. Name the missile tested by Indian scientists and defence personnel on 27.1.96 at Chandipur :

- (a) Prithvi- I (b) Prithvi-II
(c) Akash (d) PSLVD-2

U.P. U.D.A. /L.D.A. (Pre) 2006

U.P. P.C.S. (Pre)1995

Ans. (b)

Prithvi-II was first test-fired on 27 January, 1996. Successful night test-fire of Prithvi-II was conducted on 6 October, 2018. This was tested by SFC (Strategic Forces Command) at Chandipur. Prithvi-II is a single stage liquid fuelled missile.

56. Which of the following fighter aircraft was flown by former President Abdul Kalam ?

- (a) F- 16 (b) MI - 30
(c) Jaguar (d) Sukhoi - 30 MKI

R.A.S./R.T.S. (Pre) 2007

Ans. (d)

On 8 June 2006, Sukhoi-30 MKI fighter aircraft was flown by former President Abdul Kalam. His co-pilot was wing commander Ajay Rathore. Former President Pratibha Patil also flown on Sukhoi-30 on 25 November, 2009. With this, she became the first Indian woman to fly on a fighter plane.

57. To whom America is put on hold for two years under missile technology control Regime –

- (a) Indian Nuclear Organization
(b) Indian Space Research Organization.
(c) Indian Ocean Development Institute
(d) Bhabha Atomic Research Centre.

M.P.P.C.S. (Pre) 1992

Ans. (b)

On May, 1992 America imposed a ban on ISRO under missile technology control regime.

58. Among the following, who is called the founder of fully indigenous Indian missile programme?

- (a) Dr. Raja Ramanna (b) Dr. U.S. Rao
(c) Prof. M.G.K. Menon (d) Dr. Abdul Kalam

R.A.S./R.T.S. (Pre) 1993

Ans. (d)

Avul Pakir Jainulabdeen Abdul Kalam was an Indian Scientist and administrator, who also served as the 11th president of India from 2002 to 2007. He played a vital role in the development of Integrated Guided Missile Development Programme (IGMDP) in India. Nag, Agni, Prithvi, Trishul and Akash missiles are developed under this program. For his significant contribution in this field, he was awarded 'Bharat Ratna' in 1997. He is famously regarded as 'Missile Man of India'.

59. 'Shaurya' which has been successfully tested by the Defence Research and Development Organization on November 12, 2008, is a –

- (a) Surface to air ballistic missile
(b) Surface to water ballistic missile
(c) Surface to surface ballistic missile
(d) None of the above

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (c)

Shaurya is a short-range canister-launched hypersonic surface to surface tactical ballistic missile being developed by DRDO's Advanced Systems Laboratory at Hyderabad. The maximum range is estimated to be 700 km along with 1000 kg of Payload. It was successfully test-fired from the integrated test range at Chandipur, Odisha on 24 September, 2011.

60. Which one of the following statement is not correct about 'Shaurya' ?

- (a) It is a Hypersonic Missile
- (b) It is a Surface to Surface Missile
- (c) It has a Striking Range of 900 km
- (d) It was Successfully Test Fired on 24 September 2011

U.P.P.C.S. (Mains) 2011

Ans. (c)

See the explanation of above question.

61. In the context of Indian defence, consider the following statements :

1. The Shaurya missile flies with a speed of more than 8 Mach.
2. The range of Shaurya missile is more than 1600 km.

Which of the statement given above is/are correct?

- (a) Only 1
- (b) only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2009

Ans. (d)

Shaurya missile has a maximum speed of 7.5 Mach & it has a range of 700 km along with 1000 kg of payload. It is a short range ballistic missile. Hence, both given statements are correct.

62. Tejas, developed with indigenous technologies is –

- (a) A submarine
- (b) A medium-range missile
- (c) A light-combat aircraft
- (d) An aircraft carrier

U.P.P.C.S. (Mains) 2004

Ans. (c)

Indigenously developed LCA Tejas is a 4.5th generation multi-role light combat aircraft. It was named by former Prime Minister Atal Bihari Vajpayee.

63. Which one of the following is an indigenously built light combat aircraft ?

- (a) Aakash
- (b) Tejas
- (c) Vikrant
- (d) Arjun

U.P.P.C.S. (Mains) 2013

Ans. (b)

See the explanation of above question.

64. Name of the light combat aircraft developed by India :

- (a) Brahmos
- (b) Astra
- (c) Chetak
- (d) Tejas

Chhattisgarh P.C.S. (Pre) 2018

Ans. (d)

See the explanation of above question.

65. The first indigenous fighting plane of Indian Air Force is :

- (a) Tejas
- (b) Vinashak
- (c) Ojas
- (d) Agni
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

Tejas is the first indigenous fighter plane of Indian Air Force. India's Light Combat Aircraft (LCA) together with its variants, is the smallest and lightest multi-role supersonic fighter aircraft of its class. This single engine, compound-Delta-Wing Tailless Aircraft is designed and developed by ADA with HAL as the principal partner along with DRDO, BEL, CSIR, DGAQA, IAF and IN.

66. What is Tejas?

- (a) First Indian made fighter plane
- (b) Air to air missile
- (c) Remote operated aircraft
- (d) Fastest Aeroplane

M.P.P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

67. Admiral Gorshkov :

- (a) Russian Navy Cheif
- (b) Naval aircraft carrier ship
- (c) Air Force headquarter
- (d) Naval Organization

U.P.P.C.S. (Pre) 2012

Ans. (b)

Admiral Gorshkov is a Russian Naval aircraft carrier ship. This giant ship of Soviet Union served the Russian Navy from 1987 to 1996. In January 2004, India and Russia signed a \$ 947 million deal to refurbish or renovate and convert the Soviet/Russian Admiral Gorshkov into a full carrier, to be re-named INS Vikramaditya. On 16 November, 2013 it was inducted into Indian Navy.

68. Which of the following Scorpene Submarines was commissioned into Indian Navy in January, 2023?

- (a) INS Karanj (b) INS Vela
(c) INS Vagir (d) INS Kalvari

U.P. P.C.S. (Pre) 2023

Ans. (c)

Indian Navy's fifth stealth Scorpene class Submarine INS Vagir was commissioned into the Indian Navy on 23 January, 2023 at the Naval Dockyard, Mumbai in the presence of Admiral R. Hari Kumar, then Chief of the Naval Staff was the Chief Guest for the ceremony. The sixth and last Scorpene class Submarine INS Vaghsheer was commissioned into the Indian Navy on 15 January, 2025.

69. 'Sarthak' launched in August 2020 by Ministry of Defence, Government of India relates to which of the following?

- (a) Smart Bomb (b) Rocket Launcher
(c) Offshore Patrol Vessel (d) Light Combat Aircraft

U.P. P.C.S. (Pre) 2020

Ans. (c)

'Sarthak' is an offshore patrol vessel (OPV) for the Indian Coast Guard, which was launched on 13th August 2020. This vessel has been designed and built indigenously by Goa Shipyard Limited. This OPV is fitted with state-of-the-art navigation and communication equipment, sensor and machinery. The 105-metre ship displaces approximately 2,350 tonnes and is propelled by two 9,100 KW diesel engines designed to attain a maximum speed of 26 knots, with an endurance of 6,000 nautical miles.

70. On March, 2010 an offshore patrol vessel has been included into the Coast Guard group. Its name is –

- (a) Eagle (b) Vishwast
(c) Jatayu (d) Ashwast

U.P. Lower Sub. (Pre) 2008

Ans. (b)

'Vishwast', the first of the new class of offshore patrol vessel indigenously designed and built by Goa Shipyard Ltd. was inducted into Indian Coast Guard on 17 March, 2010.

71. INS-Shakti is India's first indigenously built :

- (a) Warship (b) Submarine
(c) Destroyer (d) Rocket launcher

U.P.P.C.S. (Mains) 2010

Ans. (*)

INS-SHAKTI is the second fleet tanker built by the Italian yard for the Indian Navy. It was launched on 11 October, 2010 and commissioned on 1 October, 2011. It was built by Fincantieri, an Italian shipbuilding company based in Trieste, not by India.

72. Which one of the following is the best description of 'INS Astradharini', that was in the news recently?

- (a) Amphibious warfare ship
(b) Nuclear-powered submarine
(c) Torpedo launch and recovery vessel
(d) Nuclear-powered aircraft carrier

I.A.S. (Pre) 2016

Ans. (c)

Indian Navy's first totally indigenously designed and built torpedo launch and recovery vessel INS Astradharini was commissioned at Visakhapatnam on 6th October, 2015. A 50-metre twin-hulled warship is also the first to be built at a private yard with armament fixtures. The unique design of the ship was a collaborative effort of Naval Science and Technological Laboratory, IIT Kharagpur, and Shoft Shipyard.

73. Consider the following statements :

1. Ballistic missiles are jet-propelled at subsonic speeds throughout their flights, while cruise missiles are rocket-powered only in the initial phase of flight.
2. Agni-V is a medium-range supersonic cruise missile, while BrahMos is a solid-fuelled intercontinental ballistic missile.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2023

Ans. (d)

Cruise missiles are jet-propelled at subsonic, supersonic or hypersonic speeds throughout their flights, while ballistic missiles are rocket-powered only in the initial (boost) phase of flight, after which they follow an arcing trajectory to the target. Hence, statement 1 is incorrect. BrahMos is a medium-range supersonic cruise missile, while Agni-V is a solid-fuelled intercontinental ballistic missile. Hence, statement 2 is also incorrect.

74. Missile 'BrahMos' was developed jointly by :

- (a) India and U.S.A. (b) India and Russia
(c) India and Israel (d) India and Great Britain

U.P.P.C.S. (Mains) 2004

U.P.P.C.S. (Pre) 2003

Ans. (b)

A supersonic cruise missile 'BrahMos' is a joint venture between the Russian Federation's NPO Mashinostroyeniya and DRDO. The name BrahMos is formed from the names of two rivers, the Brahmaputra of India and the Moskva of Russia. It has a speed of 2.8 to 3.0 Mach, with a original range of 290 Km. The submarine variant of BrahMos was test-fired successfully for the first time from a submerged pontoon near Vishakhapatnam at the coast of Bay of Bengal on 20 March, 2013.

75. Modern technology like BrahMos Supersonic missile by India is jointly developed by?

- (a) The United States of America (b) Russia
(c) France (d) Japan

U.P. U.D.A./L.D.A. (Pre) 2013

Ans. (b)

See the explanation of above question.

76. Supersonic Cruise Missile BrahMos is joint venture of which of the following?

- (a) India and Russia (b) India and China
(c) India and Israel (d) None of the above

U.P. P.C.S. (Pre) 2018

Ans. (a)

See the explanation of above question.

77. BrahMos is –

- (a) An aircraft (b) A Computer Virus
(c) A missile (d) A submarine

Uttarakhand U.D.A. / L.D.A. (Pre) 2003

Ans. (c)

See the explanation of above question.

78. On 18th October 2020, India successfully test fired which of the following missiles?

- (a) BrahMos Supersonic Cruise missile, Airforce version
(b) BrahMos Supersonic Cruise missile, Naval version
(c) BrahMos Subsonic missile suitable for all three forces
(d) BrahMos Supersonic Cruise missile, Army version

U.P. R.O./A.R.O. (Mains) 2016

Ans. (b)

The Naval version of BrahMos, the supersonic cruise missile was successfully test-fired on 18th October 2020 from Indian Navy's indigenously-built stealth destroyer INS Chennai, hitting a target in the Arabian Sea. The missile successfully hit the target with pin-point accuracy after performing high-level and extremely complex manoeuvres. BrahMos as 'prime strike weapon' will ensure the warships' invincibility by engaging naval surface targets at long ranges, thus making the destroyer another lethal platform of Indian Navy.

79. Which one of the following statements about BrahMos is not correct ?

- (a) It is a supersonic cruise missile.
(b) It has been built in India in collaboration with Russia.
(c) It was successfully test fired from Bay of Bengal on 22 March, 2010.
(d) Its flight range is 400 km.

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c) & (d)

BrahMos is a supersonic cruise missile which has been built in India in collaboration with Russia. On 21 March, 2010 BrahMos missile was successfully test fired from a moving warship INS Ranvir off the Odisha coast (in Bay of Bengal). Its original flight range is 290 km. In 2019, India upgraded the missile with a new range of 450/500 km. Thus, option (c) and (d) both are not correct.

80. In March, 2009 the 'BrahMos Missile' was tested at –

- (a) Pokharan
(b) Chandipur
(c) Cape Canaveral
(d) Shri Harikota

U.P.P.C.S. (Mains) 2007

Ans. (a)

BrahMos supersonic missile with a range of 290 km was first tested in January, 2009 from Pokhran test range, Rajasthan. On 29 March, 2009 land attack version of the BrahMos Block-II was tested from a Mobile Autonomous Launcher (MAL) at Pokhran test range. Another highly maneuverable version of BrahMos supersonic cruise missile was test-fired on 9 January, 2013 from the coast of Visakhapatnam in Bay of Bengal.

81. Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : In April 2024, India delivered the BrahMos Supersonic Cruise Missile to Philippines.

Reason (R) : The Missile has a range of 490 kilometres.

Select the correct answer from the codes given below :

- (a) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) (A) is true, but (R) is false.

U.P.P.C.S. (Pre) 2024

Ans. (d)

India delivered the first batch of BrahMos supersonic cruise missiles to the Philippines in April 2024. In January 2022, Philippines concluded a \$375-million deal with India for three batteries of shore-based, anti-ship variant of the BrahMos becoming the first export customer for the joint venture missile between India and Russia. India has sent to the Philippines the second of three batches of BrahMos missiles in April 2025. BrahMos is a joint venture between DRDO and Russia's NPO Mashinostroyeniya and the missile derives its name from Brahmaputra and Moskva rivers. The missile is capable of being launched from land, sea, sub-sea and air against surface and sea-based targets and has been long inducted by the Indian armed forces. The range of the missile was originally capped at 290 km as per obligations of the Missile Technology Control Regime. Following India's entry into this club in June 2016, the range has been extended to 450/500 km and work is on to extend it to 600 km and beyond. However, the range of BrahMos missile exported to Philippines is 290 km. Hence, (A) is true, but (R) is false.

82. The name of unmanned aircraft developed by Defence Research Organization (DRDO) is –

- (a) I.N.S. Virat (b) Shalki
(c) Lakshya (d) Shukoi

Jharkhand P.C.S. (Pre) 2003

Ans. (c)

Lakshya is the first unmanned aircraft developed by India. It was developed by Aeronautical Development Establishment (ADE) of DRDO.

83. Which one of the following statements is true about Technology device called 'NETRA' ?

- (a) It is a device which permits a blind man to read.
(b) It is a device which allows the army to undertake a right reconnaissance.
(c) It is an unmanned aerial vehicle which can travel up to 300 metres high.
(d) None of the above

U.P.P.C.S. (Mains) 2009

Ans. (c)

DRDO has developed an Unmanned Aerial Vehicle (UAV) especially for Anti-terrorist and counter-insurgency operations. The 1.5 kg UAV 'Netra', is a collaborative development project between Idea Forge and DRDO's Pune-based labs (R & D). It is able to fly up to a height of 200-300 metres.

84. On 20th February, 2018 who became the first-ever Indian female to fly solo 'MIG-21 Bison'?

- (a) Avani Chaturvedi (b) Mohana Singh

(c) Bhawana Kanth

(d) Shivangi Singh

U.P. P.C.S. (Mains) 2017

Ans. (a)

Flying officer Avani Chaturvedi created history by becoming first Indian woman pilot of Indian Airforce (IAF) to complete a solo flight in MIG-21 Bison fighter aircraft on 20th February, 2018.

85. Which of the following statements is not true with regard to the F-18 Super Hornet?

- (a) It is a fighter plane, which is built by U.S. aerospace company.
(b) It can be used during both night and day.
(c) Worldwide eight countries are using it in Air Force.
(d) It is built by HAL with the help of Boeing company.

U.P.P.C.S (Pre) 2002

Ans. (c)

F-18 Super Hornet is an American fighter plane, which is built by Boeing (a U.S. aerospace company). Indian company HAL has provided some tools for it. It can be used during both day and night. Presently U.S. Navy, Royal Australian Air Force and Kuwait Air Force are using this fighter plane. Thus, statement of option (c) is not correct.

86. Which of the following statement is incorrect for Active Electronically Scanned Array Radar (AESA)?

- (a) It is not capable of tracking multiple targets.
(b) The radar features include wide band RF front end.
(c) It is solid-state active phased array fire control radar.
(d) It can be adapted for various types of fighter class of aircraft.

Rajasthan P.C.S. (Pre) 2024

Ans. (a)

Active Electronically Scanned Array (AESA) radar is a multimode, solid-state active phased array fire control radar with scalable architecture that can be adapted for various types of fighter class of aircraft. The radar features include wide band RF front end, ultra low antenna side lobes, frequency and waveform agility, jammer suppression. Low Probability of Intercept and Non Co-operative Target Recognition. It is capable of tracking multiple targets with high accuracy suitable for firing missiles with interleaved Air to Air, Air to Ground and Air to Sea modes for all terrain operation. Electronics and Radar Development Establishment (LRDE), a premier lab under DRDO has developed this state-of-art radar system which is named 'Uttam'. DRDO's Uttam AESA radar is being integrated on TEJAS MK-1. Hence, statement of option (a) is incorrect, while statements of other three options are correct.

87. What is 'Terminal High Altitude Area Defense (THAAD)', sometimes seen in the news?

- (a) An Israeli radar system
- (b) India's indigenous anti-missile programme
- (c) An American anti-missile system
- (d) A defence collaboration between Japan and South Korea

I.A.S. (Pre) 2018

Ans. (c)

Terminal High Altitude Area Defense (THAAD) is an American anti-ballistic missile defence system designed to shoot down short, medium and intermediate-range ballistic missiles in their terminal phase by intercepting with a hit to kill approach. The THAAD interceptor carries no warhead, but relies on its kinetic energy of an impact to destroy the incoming missile. A kinetic energy hit minimizes the risk of exploding conventional warhead ballistic missiles and the warhead of nuclear-tipped ballistic missile will not detonate upon a kinetic energy hit.

88. Which one of the following statements best reflects the idea behind the 'Fractional Orbital Bombardment System' often talked about in media?

- (a) A hypersonic missile is launched into space to counter the asteroid approaching the Earth and explode it in space.
- (b) A spacecraft lands on another planet after making several orbital motions.
- (c) A missile is put into a stable orbit around the Earth and deorbits over a target on the Earth.
- (d) A spacecraft moves along a comet with the same speed and places a probe on its surface.

I.A.S. (Pre) 2022

Ans. (c)

The idea behind Fractional Orbital Bombardment System (FOBS) is, a warhead delivery system (missile) is put into a stable orbit around the Earth and it deorbits over a target on the Earth. A FOBS is a warhead delivery system that uses a Low-Earth Orbit (LEO) towards its target destination. Just before reaching the target, it deorbits through a retrograde engine burn. This allows it to slow down so that its trajectory intersects with its target on the ground. The Soviet Union first developed FOBS as a nuclear weapons delivery system in the 1960s. It was one of the first Soviet efforts to use space to deliver nuclear weapons. In August 2021, China tested a weapon that combined a FOBS with a hypersonic glide vehicle (HGV).

89. Treaties of 'START-I' and 'START-II' were signed :

- (a) Between USA and USSR
- (b) Between USA and China
- (c) Between USSR and China
- (d) None of the above

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

Treaties of 'START-I' and 'START-II' were signed between USA and USSR.

90. The 'New START Treaty' was in the news. What is in this treaty?

- (a) It is a bilateral strategic nuclear arms reduction treaty between the USA and the Russian Federation.
- (b) It is a multilateral energy security cooperation treaty among the members of the East Asia Summit.
- (c) It is a treaty between the Russian Federation and the European Union for the energy security co-operation.
- (d) It is a multilateral co-operation treaty among the BRICS countries for the promotion of trade.

I.A.S. (Pre) 2011

Ans. (a)

The 'New START' (START : Strategic Arms Reduction Treaty) is the treaty between the United States of America and the Russian Federation on measures for the further reduction and limitation of strategic offensive arms. It was signed on 8 April, 2010 in Prague by Former Russian president Dmitry Medvedev and then American President Barack Obama. Under the terms of the treaty, the number of strategic missile launchers will be reduced by half and it limits the number of deployed strategic nuclear warheads to 1550. It came into force on 5 February, 2011. It will now expire in February, 2026, having been extended in 2021. On 21 February, 2023, Russia suspended its participation in New START. However, it did not withdraw from the treaty, and clarified that it would continue to abide by the numerical limits in the treaty.

91. Which of the following countries has recently joined the Hague Code of Conduct (HCoC), a global initiative to prevent ballistic missile proliferation?

- (a) Palestine
- (b) Israel
- (c) India
- (d) Iran
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

On 1 June 2016, India joined the Hague code of conduct (HCoC) against ballistic missile proliferation by notifying the central contact of Hague Code of Conduct, Vienna (Austria). At present, the number of countries in the Hague Code of Conduct has reached 145.

92. Recently, USA decided to support India's membership in multilateral export control regimes called the 'Australia Group' and the 'Wassenaar Arrangement'. What is the difference between them?

1. The 'Australia Group' is an informal arrangement which aims to allow exporting countries to minimize the risk of assisting chemical and biological weapons proliferation, whereas the 'Wassenaar Arrangement' is a formal group under the OECD holding identical objectives.
2. The 'Australia Group' comprises predominantly of Asian, African and North American countries whereas the member countries of 'Wassenaar Arrangement' are predominantly from the European Union and American continents.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2011

Ans. (d)

The Australia Group is an informal group of countries established in 1985 (after the use of chemical weapons by Iraq in 1984) to help member countries to identify those exports which need to be controlled so as not to contribute to the spread of chemical and biological weapons. Now it has 43 members including all 27 members of the European Union, UK, the European Commission, USA and Canada etc. Japan, South Korea, Turkey and India are Asian Countries in this group while it has no member from African continent.

The Wassenaar Arrangement has been established in order to contribute to regional and international security and stability by promoting transparency and greater responsibility in the transfer of conventional arms and dual-use goods and technology.

The Wassenaar Arrangement is presently composed of 42 countries. It consists mainly European Union countries (and UK, Russia, Ukraine etc.) and countries of American Continent (Japan, South Korea, Turkey and India are Asian countries in it while South Africa is the only country from African continent). Now, through this explanation it is clear that the second part of statement 1 and the first part of the Statement 2 is incorrect. Thus, the correct answer is option (d).

93. With reference to the North Atlantic Treaty Organization (NATO), which of the following statements is/are correct?

I. It has 42 member nations.

II. Its Headquarters is located at Brussels.

Select the correct answer from the options given below:

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Jharkhand P.C.S. (Pre) 2023

Ans. (b)

The North Atlantic Treaty Organization (NATO), also called the North Atlantic Alliance, is an intergovernmental military alliance of 32 member States — 30 European and 2 North American. Established in the aftermath of World War II, the organization implements the North Atlantic Treaty, signed in Washington, D.C., on 4 April, 1949. NATO is a collective security system : its independent member States agree to defend each other against attacks by third parties. NATO formed with twelve founding members and has added new members ten times, most recently when Sweden joined the alliance on 7 March, 2024. In addition, NATO currently recognizes Bosnia and Herzegovina, Georgia, and Ukraine as aspiring members. NATO's main headquarters are located in Brussels, Belgium, while NATO's military headquarters are near Mons, Belgium. Hence, only statement II is correct.

94. After remaining neutral for almost 200 years, which one of the following countries officially joined the NATO in March 2024 ?

- (a) Sweden (b) North Macedonia
(c) Iceland (d) Finland

70th B.P.S.C. (Pre) 2024

Ans. (a)

Sweden had maintained a policy of neutrality in military affairs since the Napoleonic Wars, adopting a policy of "non-alignment in peace and neutrality in war". Sweden's relations with NATO date back to 1994, when Sweden joined the Partnership for Peace. Since then, Sweden's cooperation with NATO has gradually increased. Sweden officially became the 32nd member of NATO on 7th March, 2024, after decades of neutrality, marking a significant shift in its foreign policy and security stance. Before this, in April 2023, Finland became the 31st member of NATO.

95. 'Operation Pawan' is connected with –

- (a) Establishing peace in Punjab
(b) Creating more air services in India
(c) Cleansing environmental pollutions in Indian cities
(d) Operation of IPKF in Jaffna of Sri Lanka

U.P.P.C.S. (Pre) 1998

Ans. (d)

Operation Pawan was the code name assigned to the operation by the Indian Peace Keeping Force (IPKF) to take control of Jaffna from the Liberation Tigers of Tamil Eelam (LTTE), better known as the Tamil Tigers, in late 1987 to enforce the disarmament of the LTTE as a part of the Indo-Sri Lanka accord. In brutal fighting lasting about three weeks, the IPKF took control of the Jaffna Peninsula from the LTTE at the cost of the life of 214 Indian soldiers.

96. What was the theme of DefExpo-2022, held in Gujarat?

- (a) India : The Emerging Defence Manufacturing Hub
- (b) Path to Pride
- (c) Synergy for Defence
- (d) Aatmanirbharta

69th B.P.S.C. (Pre) 2023

Ans. (b)

DefExpo is India's flagship biennial event showcase Land, Naval and Homeland Security System. The first edition of DefExpo was held in 1999. The 12th edition of DefExpo-2022 was organized from 18 to 22 October, 2022 at Gandhinagar, Gujarat. The exhibition have showcased India's emergence in the shape of alliances and joint ventures for the defence industry. The theme of DefExpo-2022 was 'Path to Pride' and the aim was to showcase the might of the domestic defence industry which is now powering 'Make in India, Make for the World' resolve of the Government and the nation at large.

97. Consider the following in respect of Indian Ocean Naval Symposium (IONS) :

1. Inaugural IONS was held in India in 2015 under the chairmanship of the Indian Navy.
2. IONS is a voluntary initiative that seeks to increase maritime co-operation among navies of the littoral states of the Indian Ocean Region.

Which of the above statements is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2017

Ans. (b)

The IONS (Indian Ocean Naval Symposium) is a regional forum of Indian Ocean littoral States launched by Indian Navy in February, 2008. It is a voluntary initiative that seeks to increase maritime co-operation among navies of littoral

states of Indian Ocean Region (IOR) by providing an open and inclusive forum for discussion of regionally relevant maritime issues. The first Indian Ocean Naval Symposium (IONS) was held between 14 to 16 February, 2008 at New Delhi. The 8th IONS was hosted by Thailand in December, 2023, and the 9th IONS will be hosted by India in end of 2025. At present, IONS has 25 member-States and 9 observer countries.

98. What is Friendship-2016?

- (a) The Joint military exercise by the soldiers of Russia and Pakistan
- (b) The Joint military exercise by the soldiers of America and India
- (c) The Joint military exercise by the soldiers of India and Afghanistan
- (d) The Joint military exercise by the soldiers of India and Kazakhstan

Jharkhand P.C.S. (Pre) 2016

Ans. (a)

Friendship (Druzhba)-2016 was a joint military exercise by the soldiers of Russia and Pakistan. The joint military exercise was organised between 24 September to 10 October, 2016 at Rattu Base and Cherat unit in central Pakistan. Around 70 Russian and 130 Pakistani soldiers took part in this exercise. The name given to the military exercise is an indication of the end of the tension between the two countries witnessed during the era of cold war. It was the first joint military exercise between Russia and Pakistan. The Druzhba exercises are held annually since 2016 alternating between Pakistan and Russia. 'Friendship (Dustlik)' is also the name given to joint tactical exercises within the framework of military cooperation between Uzbekistan and India, which were held annually since 2019. Its latest edition was conducted in April, 2024 in Termez, Uzbekistan.

99. Which one of the following countries joined India and USA in the Malabar naval exercise held in July 2017?

- (a) Japan
- (b) South Korea
- (c) France
- (d) Philippines

U.P.P.C.S. (Pre) 2017

Ans. (a)

Between 10 to 17 July, 2017 the navies of India, Japan and United States of America kicked off 2017 Malabar naval exercise near Chennai in Bay of Bengal. This was the 21st edition of this naval exercise. In 2020, the Australian Navy had also joined the Malabar Exercise, making it a quadrilateral naval exercise. The 28th edition of the multinational maritime exercise Malabar-2024 took place from 8 to 18 October, 2024 in India with Harbour Phase in Visakhapatnam, followed by the Sea Phase in Bay of Bengal.

100. The 2016 joint military exercise 'Maitree' has been conducted between India and which of the following countries?

- (a) Indonesia (b) Thailand
- (c) Malaysia (d) Maldives
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

The 2016 joint military exercise 'Maitree' was conducted between 15 to 29 July, 2016 by India and Thailand in Krabi, Thailand. The latest edition of India-Thailand joint military exercise 'Maitree' was held in July, 2024 in Tak province of Thailand.

101. Which of the following countries refused India's invitation to participate in Naval Exercise 'Milan - 2018'?

- (a) Myanmar (b) Mauritius
- (c) Oman (d) Maldives

U.P. P.C.S. (Mains) 2017

Ans. (d)

Milan is a multilateral naval exercise. The 10th edition of Milan concluded with Milan Exercise Sea (MILES) conducted from 11 to 13 March, 2018 in the Andaman Sea. The Maldives refused India's invitation to participate in Naval Exercise 'Milan-2018'. The 11th edition of Milan was held from 25 February to 4 March, 2022 in Vishakhapatnam. The 12th edition of multinational exercise 'Milan' was held in February, 2024 in Vishakhapatnam with participation of over 50 friendly countries.

102. What is SLINEX II ?

- (a) An atomic submarine
- (b) Joint Naval Exercise Conducted by India and Sri Lanka

- (c) A battleship indigenously built in India
- (d) A most Modern Army Tank

U.P.P.C.S. (Mains) 2011

Ans. (b)

SLINEX-II was the joint fleet exercise between the Indian and Sri Lankan navies, conducted in Trincomalee, Sri Lanka, during September 19 to 24, 2011. SLINEX-2024 was conducted in December, 2024 at Visakhapatnam, India.

103. The bilateral military exercise 'Harimau Shakti 2018' has been conducted between India and which country?

- (a) Indonesia (b) Malaysia
- (c) New Zealand (d) South Korea
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

The bilateral joint military exercise 'Harimau Shakti 2018' between India and Malaysia was conducted from 30 April to 13 May, 2018 in dense forests of Sengai Perdik, Hulu Langat, Malaysia. The Latest edition of India-Malaysia joint military exercise 'Harimau Shakti 2024' was conducted at Bentong Camp, Pahang District, Malaysia during December 2 to 15, 2024.

104. Which city was the host of the second edition of 'The Africa-India Field Training Exercise' AFINDEX-2023?

- (a) Kinshasa (b) Bhopal
- (c) Pune (d) Johannesburg

M.P. P.C.S. (Pre) 2022

Ans. (c)

The 2nd edition of joint military exercise 'The Africa-India Field Training Exercise (AFINDEX-2023)' was held from 16 to 29 March, 2023 at Aundh, Pune. A total of 25 nations of the African continent with 124 participants and Indian troops from the SIKH, MARATHA and MAHAR Regiments participated in the multinational exercise.

105. Which of the following statements about 'Exercise Mitra Shakti-2023' are correct?

1. This was a joint military exercise between India and Bangladesh.
2. It commenced in Aundh (Pune).

3. Joint response during counter-terrorism operations was a goal of this operation.

4. Indian Air Force was a part of this exercise.

Select the answer using the code given below:

- (a) 1, 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) 2, 3 and 4

I.A.S. (Pre) 2024

Ans. (d)

The ninth edition of Joint Military Exercise between India and Sri Lanka 'Exercise MITRA SHAKTI-2023' was conducted from 16th to 29th November 2023 in Aundh (Pune). The Indian contingent, of 120 personnel, was represented mainly by troops from the Maratha Light Infantry Regiment. The Sri Lankan side is being represented by personnel from 53 Infantry Division. 15 personnel from Indian Air Force and five personnel from Sri Lankan Air Force were also participated in the exercise. The aim of the exercise was to jointly rehearse conduct of Sub Conventional operations under Chapter VII of United Nations Charter. The Scope of the exercise included synergising joint responses during counter-terrorist operations. Hence, statement 1 is incorrect while other three statements are correct.

The 10th edition of India-Sri Lanka Joint Military Exercise MITRA SHAKTI was conducted from 12th to 25th August 2024 at Army Training School, Maduru Oya, Sri Lanka. Indian contingent comprising of 106 personnel was represented by a Battalion of Rajputana Rifles along with personnel from other arms and services. The Sri Lankan contingent was represented by personnel from Gajaba Regiment of Sri Lankan Army. Aim of this Joint Exercise was to enhance joint military capability of both sides to undertake counter insurgency operations in a Sub Conventional scenario under Chapter VII of the United Nations Mandate. The exercise focused on operations in the semi-urban environment.

106. Where was the Japan-India bilateral Maritime Exercise (JIMEX-24) held this year ?

- (a) Yokosuka, Japan
(b) Bay of Bengal
(c) Western Pacific Ocean
(d) Vishakhapatnam, India

70th B.P.S.C. (Pre) 2024

Ans. (a)

The eighth edition of the bilateral Japan-India Maritime Exercise 2024 (JIMEX-24) took place in Yokosuka, Japan in June, 2024. This naval exercise involved harbour and sea phases, with professional, sports, and social activities at the harbour, followed by joint naval war drills at sea to improve their interoperability across various operations. The Indian Navy was represented by Stealth Frigate INS Shivalik and the Japan Maritime Self-Defense Force (JMSDF) was represented by the Guided Missile Destroyer JS Yugiri.

107. The I Edition of Joint Table Top Exercise (CINBAX) between the Indian Army and the Cambodian Army was held at :

- (a) Mumbai (b) Pune
(c) Jodhpur (d) Bhubaneswar

Rajasthan P.C.S. (Pre) 2024

Ans. (b)

The 1st edition of Joint Table Top Exercise, CINBAX, between the Indian Army and the Cambodian Army was conducted from 1 to 8 December, 2024 at Foreign Training Node, Pune. Exercise CINBAX was a planning exercise aimed to wargame conduct of Joint Counter Terrorism (CT) operations under Chapter VII of the United Nations Charter. The exercise focussed on discussions pertaining to establishment of Joint Training Task Force for Intelligence, Surveillance and Reconnaissance besides planning of operations in CT environment.

108. Which of the following pairs of country and joint military exercise conducted with India is incorrect?

- (a) Russia – Indra
(b) Kyrgyzstan – Khanjar
(c) Seychelles – Lamitiye
(d) Thailand – Yudh Abhyas

M.P. P.C.S. (Pre) 2025

Ans. (d)

The correctly matched pairs are as follows :

Joint Military Exercise	Participating Nations
Indra	– India and Russia
Khanjar	– India and Kyrgyzstan
Lamitiye	– India and Seychelles
Yudh Abhyas	– India and USA
Maitree	– India and Thailand

Hence, pair of option (c) is incorrect